Sanjana Umesh Sawant

Ready to relocate | email: sanjanaumesh.sawant@rutgers.edu | LinkedIn: www.linkedin.com/in/sanjana-umesh-sawant-oct082002

GitHub: https://github.com/sanjanaumeshsawant0810 | Portfolio website link: https://portfolio-sanjanasawant0810.streamlit.app

PROFESSIONAL SUMMARY

Data scientist with experience in predictive modeling, KPI analytics, SQL pipelines, survey-data transformation, and stakeholder reporting. Strongest in Python, SQL, R, PostgreSQL, scikit-learn, FastAPI, and Databricks, with work across imbalanced classification, retrieval evaluation, product analytics, and messy real-world data.

EDUCATION

Rutgers University, New Jersey, USA | M.S. in Data Science | GPA: 3.6/4.0 — May 2026

Relevant coursework: Machine Learning, Deep Learning, Large Language Models, Agentic AI, Regression and Time Series, Database Management Systems

SRM Institute of Science and Technology, Chennai, India | B.Tech. in Electronics and Computer Engineering | GPA: 9.01/10 — Jun 2024

EXPERIENCE

Data Science Intern | Helius Health AI

Remote | Mar - Ongoing 2026

- Built KPI pipelines on **Supabase(PostgreSQL)** to track onboarding, assessment completion, engagement, retention, conversion, and specialty-matching trends for leadership and sponsor reporting.
- Analyzed completion patterns across student, graduate, licensed, and unlicensed cohorts to find drop-off points, profile gaps, and re-engagement opportunities.
- Validated chatbot-captured data, reviewed database functions, and turned raw behavioral and assessment records into reusable reports and metric definitions.
- Analyzed onboarding behavior across **150+** users and found that **79.1%** completed the assessment workflow and received recommendations, helping surface workflow and data-capture integrity gaps in an AI-enabled healthcare product.
- Developed cohort segmentation and drop-off analyses across student, graduate, nurse, licensed, and unlicensed user groups, showing an **18.2%** decrease in assessment drop-off after **Segment 1**.
- Started a personality-analysis model using **PCA**-based feature transformations, reaching roughly **50% to 60%** accuracy so far; treated results as ongoing research rather than production-ready performance.

AI Agent Developer Intern | NeuralSeek

Remote | Feb - Mar 2026

- Analyzed user feedback and recommendation behavior to identify what improved trust, relevance, and decision quality in an AI recommendation workflow.
- Tested the recommendation experience with users and improved willingness to use AI-generated recommendations from **17% to 75.6%** through iteration.

PLC Programming Engineering Intern | Superpack Packaging Machines

Hyderabad, India | Oct 2023 - Jul 2024

- Improved industrial filling-machine efficiency by **4% to 8%** by testing control-program changes, debugging process issues, and working with engineers on implementation.

PROJECTS

Enterprise Knowledge Ops Copilot

Aug 2026

- Built a citation-backed enterprise document QA MVP in Python, FastAPI, Streamlit, and LangGraph over a **30-document** support corpus.
- Implemented ingestion, lexical retrieval, answer generation, and benchmark logging, reaching **96%** answer accuracy and 100% citation hit rate on a 25-question local evaluation.

Loan Default Prediction

Ongoing 2026

- Built an end-to-end credit risk modeling pipeline on **307K+** borrower records using Python, pandas, scikit-learn, and XGBoost to predict default from application-time data.
- Compared logistic regression and XGBoost using ROC-AUC, PR-AUC, calibration, and threshold analysis to connect model performance with lending decision trade-offs.

PubMedQA RAG vs CAG vs RAG+CAG

Summer 2026

- Built a biomedical evidence QA system on PubMedQA to compare RAG, CAG, and a RAG+CAG hybrid under a shared evaluation setup.
- Developed a FastAPI and Streamlit workflow that compared verdict correctness, citation match, grounding status, and latency across retrieval strategies. On the latest reported run, RAG reached **91.7%** verdict accuracy.

Effect of Use of Technology on Mental Health

Spring 2025

- Processed 200+ page non-machine-readable codebooks in Python and built an R/Databricks transformation pipeline to standardize cross-year survey variables for demographic trend analysis.
- Analyzed technology use and mental health patterns across age, gender, and income groups while keeping correlation and causation separate in the final reporting.
- Combined **NTIA** Internet Use Survey and **NSDUH** mental-health data into analysis-ready tables and identified roughly **10%** higher reported distress among women aged **20 to 35** with increased technology use.

SKILLS

Programming: Python, SQL, R, C++, MATLAB | Data Science: pandas, scikit-learn, XGBoost, statistical analysis, segmentation, model evaluation, threshold tuning, causal inference | Data & Platforms: PostgreSQL, Supabase, Databricks, PySpark, SQLite, MySQL, BigQuery, Power BI, JSONL | Engineering: FastAPI, Flask, Streamlit, Docker, Git, REST APIs, LangGraph